Dirac Equation

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Study Notes on Quantum Mechanics by Mandl [Weber Study Notes][1][2][3]

Contents

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1 Wave Function

1.1 Background

A plane wave can be represented as

$$u(x,t) = a \exp i \left(\frac{2\pi}{\lambda} x - \omega t \right) \tag{1}$$

We have the relationship between λ, ω and p, E

$$E = hv = \hbar\omega$$

$$p = \frac{h}{\lambda} = \frac{2\pi}{\lambda}\hbar$$

$$u(x,t) = a \exp \frac{i}{\hbar}(px - Et) \tag{2}$$

We see the glimpse of operators i.e.

$$i\hbar \frac{\partial u}{\partial t} = Eu$$

and

$$-i\hbar \frac{\partial u}{\partial x} = pu$$

Also from

$$-\hbar^2 \frac{\partial^2 u}{\partial x^2} = p^2 u$$

and

$$E = \frac{p^2}{2m}$$

we see the glimpse of a wave equation

$$i\hbar \frac{\partial u}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 u}{\partial x^2}$$

or for a free particle in an energy eigenstate

$$E\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

This leads to Schrödinger's equation

$$i\hbar \frac{\partial u}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 u + Vu \tag{3}$$

What we do essentially is take the classical concepts of position, momentum, angular momentum, energy etc. and replace them by their quantum mechanical operator equivalents. Then we apply those operators to a wavefunction.

1.1.1 Energy Operator

$$i\hbar\frac{\partial}{\partial t} \tag{4}$$

1.1.2 Momentum Operator

$$-i\hbar\frac{\partial}{dx} \tag{5}$$

The momentum of a particle is determined by the wave function's wavelength.

In the videos the visualisation of the wave function is as a spiral, corkscrew shape but using the x axis helps us visualise the momentum operator $-i\hbar\frac{\partial\psi}{dx}$. The key thing is the $-i$ term has the effect of rotating ψ by -90°. A corkscrew shape also means $-i\hbar\frac{\partial\psi}{dx}$ is is a constant multiplier of ψ so is therefore an eigenfunction. So we know for definite the momentum in this case of a corkscrew wavefunction. For a corkscrew wavefunction however we have no knowledge of the position as the amplitude of the wave function is the same everywhere.

1.1.3 Position Operator

The amplitude of the wavefunction at a given point determines the probability of us finding the particle at that point.

To know the position exactly we would have to have the wavefunction zero everywhere except for one value of x. But to get a wave function of that shape we'd have to superimpose many different eigenvalues of many different wavelengths of the corkscrew wave function above. This is because a corkscrew wave function which is more tightly wound has a larger value of $-i\hbar\frac{\partial\psi}{dx}$ and vice versa. We'd have to superimpose eigenfunctions corresponding to a huge number of momenta in order to specify the position accurately - but then we'd be uncertain of the momentum.

1.2 Energy Eigenfunctions

What we do is solve Schrödinger Equation 3 to find *energy* eigenfunctions $\phi(x)$ and eigenvalues. (page 31 Mandl).

For a system in a definite energy state E the following function is an eigenvector of the energy operator $i\hbar\frac{\partial}{\partial t}$

$$u(\vec{r},t) = \psi(\vec{r})\, e^{\frac{-iEt}{\hbar}}$$

From $i\hbar\frac{\partial u}{\partial t}$

$$i\hbar\frac{\partial\left(\psi(\vec{r})\, e^{\frac{-iEt}{\hbar}}\right)}{\partial t} = \psi(\vec{r})\,(i\hbar)\left(\frac{-iE}{\hbar}\right)e^{\frac{-iEt}{\hbar}} = E\left(\psi(\vec{r})\, e^{\frac{-iEt}{\hbar}}\right)$$

From $-\frac{\hbar^2}{2m}\nabla^2 u + Vu = i\hbar\frac{\partial u}{\partial t}$

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r})\, e^{\frac{-iEt}{\hbar}} + V\psi(\vec{r})\, e^{\frac{-iEt}{\hbar}} = E\psi(\vec{r})\, e^{\frac{-iEt}{\hbar}}$$

$$H\psi \equiv \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi = Eu$$

We find the solution of this ψ and it is the eigenfunction corresponding to energy state E.

ψ will also be the eigenfunction for observables l_z and l^2 in the case of the hydrogen atom (page 54 Mandl).

$\psi(\vec{r})$ also relates to the probability of finding the particle at $\vec{r}$.

1.3 Worked Example

From page 67

$$\psi_l^m(\vec{r}) = R(r)\,\Upsilon_l^m(\theta,\phi)$$

$$\psi_l(\vec{r}) = R(r)\,\frac{1}{\sqrt{2l+1}}\sum_{m=-1}^{l}\Upsilon_l^m(\theta,\phi)$$

From page 52

$$\Upsilon_l^m(\theta,\phi) = \mathcal{N}_l^m\mathcal{P}_l^m(\cos\theta)\,e^{im\phi}$$

From the operator definitions on page 54

$$l_z = -i\hbar\frac{\partial}{\partial\phi}$$

and

$$l^2 = -\hbar^2\left\{\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right\}$$

so ψ_l^m is an eigenfunction of l_z

$$l_z\psi_l^m = -i\hbar\frac{\partial}{\partial\phi}\left(R(r)\,\mathcal{N}_l^m\mathcal{P}_l^m(\cos\theta)\,e^{im\phi}\right) = m\hbar\left(R(r)\,\mathcal{N}_l^m\mathcal{P}_l^m(\cos\theta)\,e^{im\phi}\right) = \hbar m\psi_l^m$$

and also is an eigenfunction of l^2

$$l^2\psi_l^m = -\hbar^2\left\{\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right\}\left(R(r)\,\mathcal{N}_l^m\mathcal{P}_l^m(\cos\theta)\,e^{im\phi}\right)$$

and (from top of page 55). See $\left\{\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial S}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 S}{\partial\phi^2}\right\} + l(l+1)\,S = 0$. S is the spherical harmonic.

$$l^2\psi_l^m = \hbar^2\left[l(l+1)\right]\left(R(r)\,\mathcal{N}_l^m\mathcal{P}_l^m(\cos\theta)\,e^{im\phi}\right) = \hbar^2\left[l(l+1)\right]\psi_l^m$$

Also for $l = 1$ $c_n = \frac{1}{\sqrt{3}}$ so the probability of each of the 3 m states $[-1, 0, 1]$ is the same $c_n^2 = \frac{1}{3}$.

$$\psi_{l=1}(\vec{r}) = R(r)\frac{1}{\sqrt{3}}\sum_{m=-1}^{1}\Upsilon_l^m(\theta, \phi)$$

1.4 Operators

Operators corresponding to observables possess a complete set of eigenfunctions and corresponding eigenvalues (Mandl pages 12 and 62)

$$A\phi_n = a_n\phi_n$$

We can express a wavefunction as

$$\psi = \sum_{n=1}^{\infty} c_n\phi_n$$

The eigenfunctions are orthonormal

$$(\phi_m, \phi_n) = \delta_{mn}$$

Although the wave function is a scalar, it is a complex number so we can think of it in e.g. the Re, Im, x dimensions.

See these 2 videos for an excellent visualisation

Quantum Operators: `https://www.youtube.com/watch?v=LZie2QC5Jbc`

Wave Function Visualisation: `https://www.youtube.com/watch?v=KKr91v7yLcM`

If a wave function is a function of x i.e. $\psi(x)$ then x is always real but ψ may be complex e.g. e^{ix}. This can be envisaged as a spiral or helix with the real and imaginary components as 2 of the axes in the complex plane and x as the 3rd axis.

An example of a complete set of orthonormal functions that span the space of all functions of period 2π is

$$\phi_n(x) = \frac{1}{\sqrt{2\pi}}e^{inx}$$

The inner product for complex functions is defined as

$$(g(x), f(x)) \equiv \int g\bar{(x)}f(x)\,dx$$

$$(\phi_m, \phi_n) = \frac{1}{2\pi}\int_0^{2\pi} e^{i(n-m)x}dx = \delta_{nm}$$

1.5 Expectation Value

uses $(\phi_i, \phi_j) = \delta_{ij}$.

$$\psi = \sum_n c_n\phi_n$$

1.5.1 Amplitude

The coefficient c_m is called the amplitude for the wavefunction ψ to be found in the eigenstate ϕ_m. It is a complex number.

We find c_m from (ϕ_m, ψ). $|c_m|^2$ is the probability of being in the eigenstate ϕ_n

$$(\phi_m, \psi) = \int \phi_m \left(\sum c_n \phi_n\right) = c_m$$

In the same manner we can find $\psi' = \hat{A}\psi$ i.e. a new state after applying an operator. $(\phi_m, \psi') = \left(\phi_m, \hat{A}\psi\right)$

$c_m a_m$ is the amplitude for the wavefunction ψ' to be found in the eigenstate ϕ_m

$$\left(\phi_m, \hat{A}\psi\right) = \int \phi_m \left(\hat{A}\sum_n c_n \phi_n\right) = \int \phi_m \left(\sum_n c_n a_n \phi_n\right) = c_m a_m$$

1.5.2 Norm

$$(\psi, \psi) = \int \left(\sum_n c_n \phi_n\right)\left(\sum_n c_n \phi_n\right) = \sum_n |c_n|^2 = 1$$

For the observable operator $\hat{A}$ corresponding to the eigenvalue/measurement a_n, the expected value of a is given by

$$\left\langle \hat{A} \right\rangle = \left(\psi, \hat{A}\psi\right) = \int \left(\sum_n c_n \phi_n\right)\left(\hat{A}\sum_n c_n \phi_n\right) = \left(\sum_n c_n\right)\left(\sum_n c_n a_n\right) = \sum_n |c_n|^2 a_n$$

1.6 Commuting Operators

Consider 2 operators A and B. They commute if

$$AB\psi = BA\psi$$

which implies that to be true for any ψ (page 77)

$$[A, B] = AB - BA = 0$$

which is why we can measure these 3 observables

$$[H, l_z] = \left[H, l^2\right] = \left[l^2, l_z\right] = 0$$

Look to Grok for proofs of the commutation of operators. They can be quite involved to calculate so need to be methodical when calculating these.

1.6.1 More Detail

Be careful of the position operator.

Consider $[L_x, y]$

$$L_x = yp_z - zp_y$$

Note we expand out the commutator first. We don't immediately go to $[L_x, y] = L_x y - yL_x$

i.e. after the expansion we still have $[..., ...]$ expressions.

$$[L_x, y] = [(yp_z - zp_y), y] = [yp_z, y] - [zp_y, y]$$

1.6.2 First Term

The first term using $[AB, C] = A[B, C] + [A, C]B$

$$[yp_z, y] = y[p_z, y] + [y, y]p_z = y\left[-i\hbar\frac{\partial}{\partial z}, y\right] + 0 = 0$$

We didn't need to expand $[A, B] = AB - BA$ to get this result.

1.6.3 Second Term

$$\begin{aligned}[zp_y, y] &= z[p_y, y] + [z, y]p_y \\ &= z\left[-i\hbar\frac{\partial}{\partial y}, y\right] + 0\end{aligned}$$

Now we finally expand out $\left[-i\hbar\frac{\partial}{\partial y}, y\right]$ applied to a test function f . Remember the position operator y operates on a function f by multiplying that function by y

$$[p_y, y] = p_y(yf) - y(p_y f)$$

$$\begin{aligned}\left[-i\hbar\frac{\partial}{\partial y}, y\right] &= \left(-i\hbar\frac{\partial}{\partial y}(yf)\right) - y\left(-i\hbar\frac{\partial}{\partial y}(f)\right) \\ &= \left(-i\hbar\frac{\partial}{\partial y}(yf)\right) + \left(i\hbar y\frac{\partial f}{\partial y}\right) \\ &= \left(-i\hbar f\frac{\partial y}{\partial y}\right) + \left(-i\hbar y\frac{\partial f}{\partial y}\right) + \left(i\hbar y\frac{\partial f}{\partial y}\right) = -i\hbar f\end{aligned}$$

So we are left with from $[A, B]f = (AB - BA)f$

$$[L_x, y] = 0 - (-i\hbar z)$$

$$[L_x, y] = i\hbar z$$

2 Dirac Equation

Dirac's equation (which we will show the origin of in this section) is

$$\left[\gamma_\mu \frac{\partial}{\partial x_\mu} + M\right]\psi = 0$$

The Klein-Gordon equation below is second order in time $\frac{\partial^2}{\partial t^2}$. Dirac was looking for first order in time. The resulting wavefunction ψ, which has 4 components, explains spin-up and spin-down and also particles and anti-particles and the magnetic moment.

2.1 Klein-Gordon Equation

The Klein-Gordon equation is

$$\left[\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{m^2c^2}{\hbar^2}\right]\psi = 0$$

This follows from

$$E^2 = m^2c^4 + c^2p^2$$

Replacing the classical energy and momentum in $E^2 = m^2c^4 + c^2p^2$ with operators $i\hbar\frac{\partial}{\partial t}$ and $-i\hbar\nabla$.

$$\left[-\hbar^2\frac{\partial^2}{\partial t^2} - m^2c^4 + c^2\hbar^2\nabla^2\right]\psi = 0$$

$$\left[\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{m^2c^2}{\hbar^2}\right]\psi = 0 \tag{6}$$

2.1.1 Low Momentum Limit

In the limit of low velocity/momentum

$$\frac{E^2}{m^2c^4} = 1 + \frac{p^2}{m^2c^2}$$

$$\frac{E}{mc^2} = \left(1 + \frac{p^2}{m^2c^2}\right)^{\frac{1}{2}} \approx 1 + \frac{p^2}{2m^2c^2}$$

$$E = mc^2 + \frac{p^2}{2m}$$

2.2 4-Vector Notation

The d'Alembertian (wave operator) is defined as

$$\Box \equiv \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} = \frac{\partial}{\partial x_\mu}\frac{\partial}{\partial x^\mu}$$

where μ ranges from 0 to 3. Equivalent to vector space (ict, x, y, z) because $\left(\frac{\partial}{ic\partial t}\right)^2 = \left(-\frac{i\partial}{c\partial t}\right)^2 = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}$

$M \equiv \frac{mc}{\hbar}$ so $\left[\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{m^2c^2}{\hbar^2}\right]\psi = 0$ becomes

$$\left[\Box - M^2\right]\psi = 0$$

$$\left[\frac{\partial}{\partial x_\mu}\frac{\partial}{\partial x^\mu} - M^2\right]\psi = 0 \tag{7}$$

2.3 First Order Equation

Dirac starts out by formulating a first order equation which is $= 0$

$$\left[\gamma_\mu\frac{\partial}{\partial x_\mu} + M\right]\psi = 0 \tag{8}$$

To find the meaning of γ_μ we the apply the operator $\left(\gamma_\lambda\frac{\partial}{\partial x_\lambda} - M\right)$ to this

$$\left(\gamma_\lambda\frac{\partial}{\partial x_\lambda} - M\right)\left(\gamma_\mu\frac{\partial}{\partial x_\mu} + M\right)$$

Explicitly

$$\left(\gamma_0\frac{\partial}{\partial x_0} + \gamma_1\frac{\partial}{\partial x_1} + \gamma_2\frac{\partial}{\partial x_2} + \gamma_3\frac{\partial}{\partial x_3}\right)\left(\gamma_0\frac{\partial}{\partial x_0} + \gamma_1\frac{\partial}{\partial x_1} + \gamma_2\frac{\partial}{\partial x_2} + \gamma_3\frac{\partial}{\partial x_3}\right) - M^2$$

$$\begin{aligned}
&\gamma_0\frac{\partial}{\partial x_0}\left(\gamma_0\frac{\partial}{\partial x_0} + \gamma_1\frac{\partial}{\partial x_1} + \gamma_2\frac{\partial}{\partial x_2} + \gamma_3\frac{\partial}{\partial x_3}\right) + \\
&\gamma_1\frac{\partial}{\partial x_1}\left(\gamma_0\frac{\partial}{\partial x_0} + \gamma_1\frac{\partial}{\partial x_1} + \gamma_2\frac{\partial}{\partial x_2} + \gamma_3\frac{\partial}{\partial x_3}\right) + \\
&\gamma_2\frac{\partial}{\partial x_2}\left(\gamma_0\frac{\partial}{\partial x_0} + \gamma_1\frac{\partial}{\partial x_1} + \gamma_2\frac{\partial}{\partial x_2} + \gamma_3\frac{\partial}{\partial x_3}\right) + \\
&\gamma_3\frac{\partial}{\partial x_3}\left(\gamma_0\frac{\partial}{\partial x_0} + \gamma_1\frac{\partial}{\partial x_1} + \gamma_2\frac{\partial}{\partial x_2} + \gamma_3\frac{\partial}{\partial x_3}\right) - \\
&M^2
\end{aligned}$$

The Klein-Gordon equation is $\left[\frac{\partial}{\partial x_\mu}\frac{\partial}{\partial x^\mu} - M^2\right]\psi = 0.$

The above gives the $\mu = \lambda$ terms as e.g.

$$\gamma_1 \gamma_1 \frac{\partial}{\partial x_1} \frac{\partial}{\partial x_1}$$

and $\mu \neq \lambda$ terms like

$$\gamma_1 \frac{\partial}{\partial x_1} \left(\gamma_2 \frac{\partial}{\partial x_2} \right) + \gamma_2 \frac{\partial}{\partial x_2} \left(\gamma_1 \frac{\partial}{\partial x_1} \right) = (\gamma_1 \gamma_2 + \gamma_2 \gamma_1) \frac{\partial}{\partial x_1} \frac{\partial}{\partial x_2}$$

So to get the Klein-Gordon equation we need $(\gamma_\lambda \gamma_\mu + \gamma_\mu \gamma_\lambda) = 2\delta_{\lambda\mu}$. We need γ_μ and γ_λ to not commute which would indicate matrices.

2.3.1 Matrices

Using the Pauli spin matrices $\sigma_x, \sigma_y, \sigma_z$ we can create 4 matrices $\gamma_0, \gamma_1, \gamma_2, \gamma_3$ that satisfy the anticommutation relation

$$(\gamma_\lambda \gamma_\mu + \gamma_\mu \gamma_\lambda) = 2\delta_{\lambda\mu}$$

Note that we need 4 matrices. They also have to be 4×4 matrices. We couldn't just choose the 3 Pauli matrices because there is no fourth 2×2 matrix that anticommutes with all three Pauli matrices.

Mandl uses

$$\gamma_k = \begin{pmatrix} [0] & [-i\sigma_k] \\ [i\sigma_k] & [0] \end{pmatrix}, \gamma_0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$$

The appearance of the factors of i multiplying the Pauli matrices mirrors the quaternionic construction discussed in Section D.

Quaternions were originally used to calculate 3-D rotations long before quantum mechanics but they used symbolic algebraic notation i, j, k, not a matrix representation. The identifications $i \mapsto i\sigma_x, j \mapsto i\sigma_y, k \mapsto i\sigma_z$ give an explicit matrix representation of the quaternion units.

Hence the Pauli matrices could in principle have appeared earlier. (see section D.3)

These are the Weyl (chiral) representation with the standard Pauli matrices below.

$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\gamma_1 = \begin{pmatrix} [0] & [-i\sigma_x] \\ [i\sigma_x] & [0] \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}$$

$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$$\gamma_2 = \begin{pmatrix} [0] & [-i\sigma_y] \\ [i\sigma_y] & [0] \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\gamma_3 = \begin{pmatrix} [0] & [-i\sigma_z] \\ [i\sigma_z] & [0] \end{pmatrix} = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix}$$

We define γ_0

$$\gamma_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

2.3.2 Other Matrix Representation

I have seen this $\begin{pmatrix} [0] & [\sigma] \\ [-\sigma] & [0] \end{pmatrix}$ which for $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ gives

$$\gamma_1 = \begin{pmatrix} [0] & [\sigma_x] \\ [-\sigma_x] & [0] \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

But that doesn't multiply correctly $\gamma_1\gamma_1 = -\mathbb{1}$ so I stayed with Mandl version $\begin{pmatrix} [0] & [-i\sigma] \\ [i\sigma] & [0] \end{pmatrix}$

$$\gamma_1\gamma_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

2.4 Dirac Hamiltonian

From eq 8 $\left[\gamma_\mu \frac{\partial}{\partial x_\mu} + M\right]\psi = 0$ and $x_0 = ict$ we get the below where k ranges from 1 to 3

$$\left[\frac{\gamma_0}{ic}\frac{\partial}{\partial t} + \gamma_k\frac{\partial}{\partial x_k} + M\right]\psi = 0$$

$$-\frac{\gamma_0}{ic}\frac{\partial\psi}{\partial t} = \left(\gamma_k\frac{\partial}{\partial x_k} + M\right)\psi$$

Note that $\gamma_0^{-1} = \gamma_0$ i.e.

$$\gamma_0\gamma_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbb{1}$$

$$\frac{\partial\psi}{\partial t} = -ic\gamma_0 \left(\gamma_k \frac{\partial}{\partial x_k} + M\right)\psi$$

With $M \equiv \frac{mc}{\hbar}$

$$i\hbar\frac{\partial\psi}{\partial t} = \gamma_0\hbar c \left(\gamma_k \frac{\partial}{\partial x_k} + \frac{mc}{\hbar}\right)\psi$$

$$i\hbar\frac{\partial\psi}{\partial t} = \gamma_0\gamma_k c\left(\hbar\frac{\partial}{\partial x_k}\right)\psi + mc^2\gamma_0\psi = i\gamma_0\gamma_k c\left(-i\hbar\frac{\partial}{\partial x_k}\right)\psi + mc^2\gamma_0\psi$$

using $\mathbf{p} = -i\hbar\frac{\partial}{\partial x_k}$

$$i\hbar\frac{\partial\psi}{\partial t} = ic\gamma_0\gamma_k\mathbf{p}\psi + mc^2\gamma_0\psi$$

$$H = ic\gamma_0\gamma_k\mathbf{p} + mc^2\gamma_0 \tag{9}$$

2.4.1 Standard Form

The standard form of the Hamiltonian operator is usually given as

$$\hat{H} = c\alpha\cdot\mathbf{p} + \beta mc^2$$

with $\beta = \gamma_0$ and $\alpha = \begin{pmatrix} 0 & \sigma \\ \sigma & 0 \end{pmatrix}$ but that is equivalent to Eq 9 i.e. $\alpha = i\gamma_0\gamma_1$ e.g.

$$\alpha_1 = \begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$i\gamma_0\gamma_1 = i\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

2.4.2 Gamma Products

$$\gamma_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \gamma_1 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}$$

γ_0 simply multiplies the first 2 rows by $+1$ and the second 2 rows by -1. Consider $\gamma_0\gamma_1$

$$\gamma_0\gamma_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}$$

The meaning of e.g. $\gamma_0\gamma_1\frac{\partial}{\partial x}$ is shown below. It is applied to a 4 element spinor as shown

$$\gamma_0\gamma_1\frac{\partial}{\partial x_1}\psi = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{\partial\psi_1}{\partial x_1} \\ \frac{\partial\psi_2}{\partial x_1} \\ \frac{\partial\psi_3}{\partial x_1} \\ \frac{\partial\psi_4}{\partial x_1} \end{pmatrix}$$

2.5 Dirac Hamiltonian Solutions

$$i\hbar\frac{\partial\psi}{\partial t} = H\psi = \left(ic\gamma_0\gamma_k\mathbf{p} + mc^2\gamma_0\right)\psi$$

Remember $\mathbf{p} = -i\hbar\frac{\partial}{\partial x_k}$ so $\gamma_k\mathbf{p}$ is a summation over the 3 space co-ordinates.

We assume a plane wave solution for the Dirac equation as below. u will be a constant and it must be a 4 row matrix (because we are multiplying by the 4×4 γ matrices).

$$\psi = u\exp\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r} - Et) = u\exp\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r})\, e^{-\frac{iE}{\hbar}t}$$

ψ is an energy eigenstate i.e.

$$i\hbar\frac{\partial\psi}{\partial t} = (i\hbar)\left(-\frac{iE}{\hbar}\right) u\exp\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r})\, e^{-\frac{iE}{\hbar}t} = E\psi$$

so we have

$$E\psi = \left(ic\gamma_0\gamma_k\mathbf{p} + mc^2\gamma_0\right)\psi$$

with $\psi = ue^{\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r})}e^{-\frac{iE}{\hbar}t}$. Let us call $\psi_{\mathbf{p}} = ue^{\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r})}$ so $\psi = \psi_{\mathbf{p}}e^{-\frac{iE}{\hbar}t}$. The $e^{-\frac{iE}{\hbar}t}$ term cancels since $\mathbf{p}$ doesn't operate on it. We are left with

$$E\psi_{\mathbf{p}} = \left(ic\gamma_0\gamma_k\mathbf{p} + mc^2\gamma_0\right)\psi_{\mathbf{p}}$$

$\psi_{\mathbf{p}}$ is also a momentum eigenstate e.g. for the x term

$$-i\hbar\frac{\partial\psi_{\mathbf{p}}}{\partial x} = -i\hbar\frac{\partial}{\partial x}\left(u\exp\frac{i}{\hbar}\left(\mathbf{p}\cdot\mathbf{r}\right)\right) = (-i\hbar)\frac{\partial}{\partial x}\left(u\exp\frac{i}{\hbar}\left(p_x x + p_y y + p_z z\right)\right)$$

$$-i\hbar\frac{\partial\psi_{\mathbf{p}}}{\partial x} = p_x\psi_{\mathbf{p}}$$

From

$$ic\gamma_0\gamma_k\mathbf{p}\psi_{\mathbf{p}} = ic\gamma_0\gamma_k\left(-i\hbar\frac{\partial}{\partial x_k}\right)\psi_{\mathbf{p}}$$

$$= ic\gamma_0\left\{\gamma_1\left(-i\hbar\frac{\partial\psi_{\mathbf{p}}}{\partial x_1}\right) + \gamma_2\left(-i\hbar\frac{\partial\psi_{\mathbf{p}}}{\partial x_2}\right) + \gamma_3\left(-i\hbar\frac{\partial\psi_{\mathbf{p}}}{\partial x_3}\right)\right\}$$

$$= ic\gamma_0\left(\gamma_1 p_1\psi_{\mathbf{p}} + \gamma_2 p_2\psi_{\mathbf{p}} + \gamma_3 p_3\psi_{\mathbf{p}}\right)$$

$$ic\gamma_0\gamma_k\mathbf{p} = ic\gamma_0\gamma_1 p_1 + ic\gamma_0\gamma_2 p_2 + ic\gamma_0\gamma_3 p_3$$

$$\gamma_0\gamma_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma_0\gamma_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma_0\gamma_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ -i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}$$

$$ic\gamma_0\gamma_k\mathbf{p} = ic\begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}p_1 + ic\begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}p_2 + ic\begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ -i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}p_3$$

Because p is constant, i.e. eigenvalue of $\psi_{\mathbf{p}}$, we are not using the momentum operator here so we just multiply through the matrices with p_1 p_2 and p_3

$$= ic\begin{pmatrix} 0 & 0 & 0 & -ip_1 \\ 0 & 0 & -ip_1 & 0 \\ 0 & -ip_1 & 0 & 0 \\ -ip_1 & 0 & 0 & 0 \end{pmatrix} + ic\begin{pmatrix} 0 & 0 & 0 & -p_2 \\ 0 & 0 & p_2 & 0 \\ 0 & -p_2 & 0 & 0 \\ p_2 & 0 & 0 & 0 \end{pmatrix} + ic\begin{pmatrix} 0 & 0 & -ip_3 & 0 \\ 0 & 0 & 0 & ip_3 \\ -ip_3 & 0 & 0 & 0 \\ 0 & ip_3 & 0 & 0 \end{pmatrix}$$

$$mc^2\gamma_0 = \begin{pmatrix} mc^2 & 0 & 0 & 0 \\ 0 & mc^2 & 0 & 0 \\ 0 & 0 & -mc^2 & 0 \\ 0 & 0 & 0 & -mc^2 \end{pmatrix}$$

$\left(ic\gamma_0\gamma_k\mathbf{p} + mc^2\gamma_0\right)\psi_{\mathbf{p}} = E\psi_{\mathbf{p}}$. But from $\psi_{\mathbf{p}} = ue^{\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r})}$, the $e^{\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r})}$ terms cancel so we are just left with u

$$\begin{pmatrix} mc^2 & 0 & ic\left(-ip_3\right) & ic\left(-ip_1 - p_2\right) \\ 0 & mc^2 & ic\left(-ip_1 + p_2\right) & ic\left(ip_3\right) \\ ic\left(-ip_3\right) & ic\left(-ip_1 - p_2\right) & -mc^2 & 0 \\ ic\left(-ip_1 + p_2\right) & ic\left(ip_3\right) & 0 & -mc^2 \end{pmatrix} u = Eu$$

$$\begin{pmatrix} \left(mc^2 - E\right) & 0 & ic\left(-ip_3\right) & ic\left(-ip_1 - p_2\right) \\ 0 & \left(mc^2 - E\right) & ic\left(-ip_1 + p_2\right) & ic\left(ip_3\right) \\ ic\left(-ip_3\right) & ic\left(-ip_1 - p_2\right) & \left(-mc^2 - E\right) & 0 \\ ic\left(-ip_1 + p_2\right) & ic\left(ip_3\right) & 0 & \left(-mc^2 - E\right) \end{pmatrix} u = 0$$

This gives the 4 equations

$$\left(mc^2 - E\right)u_1 + cp_3u_3 + cp_1u_4 - (ic)\,p_2u_4 = 0$$

$$\left(mc^2 - E\right)u_2 + cp_1u_3 + (ic)\,p_2u_3 - cp_3u_4 = 0$$

$$cp_3u_1 + cp_1u_2 - (ic)\,p_2u_2 - \left(mc^2 + E\right)u_3 = 0$$

$$cp_1u_1 + (ic)\,p_2u_1 - cp_3u_2 - \left(mc^2 + E\right)u_4 = 0$$

$$u_1 = -\frac{cp_3u_3 + cp_1u_4 - (ic)\,p_2u_4}{(mc^2 - E)}$$

$$u_2 = -\frac{cp_1u_3 + (ic)\,p_2u_3 - cp_3u_4}{(mc^2 - E)}$$

$$u_3 = \frac{cp_3u_1 + cp_1u_2 - (ic)\,p_2u_2}{(E + mc^2)}$$

$$u_4 = \frac{cp_1u_1 + (ic)\,p_2u_1 - cp_3u_2}{(E + mc^2)}$$

These equations are coupled e.g. if I pick u_1 and u_2 then u_3 and u_4 are determined.

Substituting u_3 and u_4 in the first row

$$\left(mc^2 - E\right) u_1 + cp_3 u_3 + cp_1 u_4 - (ic)\, p_2 u_4 = 0$$

$$\begin{aligned}\left(E + mc^2\right)\left(mc^2 - E\right) u_1 + cp_3 \left(cp_3 u_1 + cp_1 u_2 - (ic)\, p_2 u_2\right) \\ +cp_1 \left(cp_1 u_1 + (ic)\, p_2 u_1 - cp_3 u_2\right) \\ - (ic)\, p_2 \left(cp_1 u_1 + (ic)\, p_2 u_1 - cp_3 u_2\right)\end{aligned}$$

$$\begin{aligned}\left(E + mc^2\right)\left(mc^2 - E\right) u_1 + \\ \left(c^2 p_3^2 u_1 + c^2 p_3 p_1 u_2 - ic^2 p_3 p_2 u_2\right) \\ \left(c^2 p_1^2 u_1 + ic^2 p_1 p_2 u_1 - c^2 p_1 p_3 u_2\right) \\ \left(-ic^2 p_2 p_1 u_1 + c^2 p_2^2 u_1 + ic^2 p_2 p_3 u_2\right)\end{aligned}$$

$$\left(E + mc^2\right)\left(mc^2 - E\right) u_1 + c^2\left(p_1^2 + p_2^2 + p_3^2\right) u_1 = 0$$

$$E = \pm\sqrt{m^2 c^4 + c^2\left(p_1^2 + p_2^2 + p_3^2\right)}$$

Similarly substituting u_3 and u_4 in the second row gives

$$\left(E + mc^2\right)\left(mc^2 - E\right) u_2 + c^2\left(p_1^2 + p_2^2 + p_3^2\right) u_2 = 0$$

If we choose

$$\phi = \left(\begin{array}{c} u_1 \\ u_2 \end{array}\right) = \left(\begin{array}{c} 1 \\ 0 \end{array}\right)$$

Note ϕ is the freely chosen part of the solution. At this point I could choose anything e.g. $\left(\begin{array}{c} 4 \\ 0 \end{array}\right)$ or $\left(\begin{array}{c} 1 \\ -1 \end{array}\right)$, there is nothing in the maths to stop me doing that.

We get the following for u_3 and u_4

$$u_3 = \frac{cp_3}{\left(E + mc^2\right)}$$

$$u_4 = \frac{c\left(p_1 + ip_2\right)}{\left(E + mc^2\right)}$$

Note in the non relativistic limit $E = mc^2$ and p is small, so $\chi \ll \phi$ where $\chi = \left(\begin{array}{c} u_3 \\ u_4 \end{array}\right)$. This is why ϕ is sometimes called the "large component" and χ the "small component".

$$\begin{pmatrix} \phi \\ \chi \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ \frac{cp_3}{(E+mc^2)} \\ \frac{c(p_1+ip_2)}{(E+mc^2)} \end{pmatrix}$$

The other independent solution for the other independent vector $\phi = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is

$$\begin{pmatrix} \phi \\ \chi \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ \frac{c(p_1-ip_2)}{(E+mc^2)} \\ -\frac{cp_3}{(E+mc^2)} \end{pmatrix}$$

These are the two linearly independent positive energy solutions of the Dirac equation i.e. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ cannot be written in terms of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

2.5.1 Negative Energy Solution

We get these simply by replacing E with $-E$ in the positive energy solutions. (E is the absolute value i.e. $|E|$) e.g.

$$\begin{pmatrix} \phi \\ \chi \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ \frac{cp_3}{(-E+mc^2)} \\ \frac{c(p_1+ip_2)}{(-E+mc^2)} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -\frac{cp_3}{(E-mc^2)} \\ -\frac{c(p_1+ip_2)}{(E-mc^2)} \end{pmatrix}$$

2.5.2 All Solutions

So there are 4 independent solutions to the Dirac equation. In the below E is the absolute value i.e. $|E|$

$$\begin{pmatrix} 1 \\ 0 \\ \frac{cp_3}{(E+mc^2)} \\ \frac{c(p_1+ip_2)}{(E+mc^2)} \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \frac{c(p_1-ip_2)}{(E+mc^2)} \\ -\frac{cp_3}{(E+mc^2)} \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -\frac{cp_3}{(E-mc^2)} \\ -\frac{c(p_1+ip_2)}{(E-mc^2)} \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -\frac{c(p_1-ip_2)}{(E-mc^2)} \\ \frac{cp_3}{(E-mc^2)} \end{pmatrix}$$

2.5.3 Spinor

$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}$ is a spinor field. The 4 components are all part of a single quantum state.

The normalised spinor for $\phi = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is

$$u_{norm} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \sqrt{\frac{E+mc^2}{2mc^2}} \begin{pmatrix} 1 \\ 0 \\ \frac{cp_3}{(E+mc^2)} \\ \frac{c(p_1+ip_2)}{(E+mc^2)} \end{pmatrix}$$

The Dirac norm is given by $\bar{u}u = u^\dagger \gamma_0 u$ i.e. not the Euclidean norm $u^\dagger u$

$$u^\dagger \gamma_0 u = \frac{E+mc^2}{2mc^2} \begin{pmatrix} 1 & 0 & \frac{cp_3}{(E+mc^2)} & \frac{c(p_1-ip_2)}{(E+mc^2)} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \frac{cp_3}{(E+mc^2)} \\ \frac{c(p_1+ip_2)}{(E+mc^2)} \end{pmatrix}$$

$$= \frac{E+mc^2}{2mc^2} \begin{pmatrix} 1 & 0 & \left(-\frac{cp_3}{(E+mc^2)}\right) & \left(-\frac{c(p_1-ip_2)}{(E+mc^2)}\right) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \frac{cp_3}{(E+mc^2)} \\ \frac{c(p_1+ip_2)}{(E+mc^2)} \end{pmatrix}$$

$$= \frac{E+mc^2}{2mc^2} \left(1 + 0 - \frac{c^2p_3^2}{(E+mc^2)^2} - \frac{c^2p_1^2 + c^2p_2^2}{(E+mc^2)^2} \right)$$

$$= \frac{E+mc^2}{2mc^2} \left(\frac{\left(E+mc^2\right)^2}{(E+mc^2)^2} - \frac{c^2p^2}{(E+mc^2)^2} \right)$$

From $E^2 = m^2c^4 + c^2p^2$

$$= \frac{E+mc^2}{2mc^2} \left(\frac{\left(E+mc^2\right)^2 - E^2 + m^2c^4}{(E+mc^2)^2} \right)$$

$$\bar{u}u = u^\dagger \gamma_0 u = \frac{1}{2mc^2} \left(\frac{2Emc^2 + 2m^2c^4}{E+mc^2} \right) = \left(\frac{E+mc^2}{E+mc^2} \right) = 1$$

2.5.4 Non Relativistic Limit

From $H = ic\gamma_0\gamma_k\mathbf{p} + mc^2\gamma_0$ and $\mathbf{p} = -i\hbar\frac{\partial}{\partial x_k}$. Unlike the previous section, we won't assume ψ is an eigenstate of momentum $\left(-i\hbar\frac{\partial\psi}{\partial x} \neq p_x\psi\right)$

$$ic\gamma_0\gamma_1 \left(-i\hbar\frac{\partial}{\partial x_1}\right)\psi = \hbar c \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{\partial\psi_1}{\partial x_1} \\ \frac{\partial\psi_2}{\partial x_1} \\ \frac{\partial\psi_3}{\partial x_1} \\ \frac{\partial\psi_4}{\partial x_1} \end{pmatrix} = hc \begin{pmatrix} -i\frac{\partial\psi_4}{\partial x_1} \\ -i\frac{\partial\psi_3}{\partial x_1} \\ -i\frac{\partial\psi_2}{\partial x_1} \\ -i\frac{\partial\psi_1}{\partial x_1} \end{pmatrix}$$

$$ic\gamma_0\gamma_2\left(-i\hbar\frac{\partial}{\partial x_2}\right)\psi=\hbar c\begin{pmatrix}0&0&0&-1\\0&0&1&0\\0&-1&0&0\\1&0&0&0\end{pmatrix}\begin{pmatrix}\frac{\partial\psi_1}{\partial x_2}\\\frac{\partial\psi_2}{\partial x_2}\\\frac{\partial\psi_3}{\partial x_2}\\\frac{\partial\psi_4}{\partial x_2}\end{pmatrix}=hc\begin{pmatrix}-\frac{\partial\psi_4}{\partial x_2}\\\frac{\partial\psi_3}{\partial x_2}\\-\frac{\partial\psi_2}{\partial x_2}\\\frac{\partial\psi_1}{\partial x_2}\end{pmatrix}$$

$$ic\gamma_0\gamma_3\left(-i\hbar\frac{\partial}{\partial x_3}\right)\psi=\hbar c\begin{pmatrix}0&0&-i&0\\0&0&0&i\\-i&0&0&0\\0&i&0&0\end{pmatrix}\begin{pmatrix}\frac{\partial\psi_1}{\partial x_3}\\\frac{\partial\psi_2}{\partial x_3}\\\frac{\partial\psi_3}{\partial x_3}\\\frac{\partial\psi_4}{\partial x_3}\end{pmatrix}=hc\begin{pmatrix}-i\frac{\partial\psi_3}{\partial x_3}\\i\frac{\partial\psi_4}{\partial x_3}\\-i\frac{\partial\psi_1}{\partial x_3}\\i\frac{\partial\psi_2}{\partial x_3}\end{pmatrix}$$

$$\left(ic\gamma_0\gamma_k\mathbf{p}+mc^2\gamma_0\right)\psi=\hbar c\left\{\begin{pmatrix}-i\frac{\partial\psi_4}{\partial x_1}\\-i\frac{\partial\psi_3}{\partial x_1}\\-i\frac{\partial\psi_2}{\partial x_1}\\-i\frac{\partial\psi_1}{\partial x_1}\end{pmatrix}+\begin{pmatrix}-\frac{\partial\psi_4}{\partial x_2}\\\frac{\partial\psi_3}{\partial x_2}\\-\frac{\partial\psi_2}{\partial x_2}\\\frac{\partial\psi_1}{\partial x_2}\end{pmatrix}+\begin{pmatrix}-i\frac{\partial\psi_3}{\partial x_3}\\i\frac{\partial\psi_4}{\partial x_3}\\-i\frac{\partial\psi_1}{\partial x_3}\\i\frac{\partial\psi_2}{\partial x_3}\end{pmatrix}\right\}+\begin{pmatrix}mc^2\psi_1\\mc^2\psi_2\\-mc^2\psi_3\\-mc^2\psi_4\end{pmatrix}$$

Let us look at the positive energy state E. For the non relativistic limit, $E\approx mc^2$. Note in taking $E\approx mc^2$ we are explicitly restricting to the positive energy solutions of the Dirac equation.

$$\begin{pmatrix}E\psi_1\\E\psi_2\\E\psi_3\\E\psi_4\end{pmatrix}=\hbar c\left\{\begin{pmatrix}-i\frac{\partial\psi_4}{\partial x_1}\\-i\frac{\partial\psi_3}{\partial x_1}\\-i\frac{\partial\psi_2}{\partial x_1}\\-i\frac{\partial\psi_1}{\partial x_1}\end{pmatrix}+\begin{pmatrix}-\frac{\partial\psi_4}{\partial x_2}\\\frac{\partial\psi_3}{\partial x_2}\\-\frac{\partial\psi_2}{\partial x_2}\\\frac{\partial\psi_1}{\partial x_2}\end{pmatrix}+\begin{pmatrix}-i\frac{\partial\psi_3}{\partial x_3}\\i\frac{\partial\psi_4}{\partial x_3}\\-i\frac{\partial\psi_1}{\partial x_3}\\i\frac{\partial\psi_2}{\partial x_3}\end{pmatrix}\right\}+\begin{pmatrix}mc^2\psi_1\\mc^2\psi_2\\-mc^2\psi_3\\-mc^2\psi_4\end{pmatrix}$$

$$\begin{pmatrix}\left(E-mc^2\right)\psi_1\\\left(E-mc^2\right)\psi_2\\2mc^2\psi_3\\2mc^2\psi_4\end{pmatrix}=\hbar c\left\{\begin{pmatrix}-i\frac{\partial\psi_4}{\partial x_1}\\-i\frac{\partial\psi_3}{\partial x_1}\\-i\frac{\partial\psi_2}{\partial x_1}\\-i\frac{\partial\psi_1}{\partial x_1}\end{pmatrix}+\begin{pmatrix}-\frac{\partial\psi_4}{\partial x_2}\\\frac{\partial\psi_3}{\partial x_2}\\-\frac{\partial\psi_2}{\partial x_2}\\\frac{\partial\psi_1}{\partial x_2}\end{pmatrix}+\begin{pmatrix}-i\frac{\partial\psi_3}{\partial x_3}\\i\frac{\partial\psi_4}{\partial x_3}\\-i\frac{\partial\psi_1}{\partial x_3}\\i\frac{\partial\psi_2}{\partial x_3}\end{pmatrix}\right\}$$

The first row becomes

$$\begin{aligned}\left(E-mc^2\right)\psi_1=&\frac{(\hbar c)^2}{2mc^2}\left\{-i\frac{\partial}{\partial x_1}\left(-i\frac{\partial\psi_1}{\partial x_1}+\frac{\partial\psi_1}{\partial x_2}+i\frac{\partial\psi_2}{\partial x_3}\right)\right\}\\&+\frac{(\hbar c)^2}{2mc^2}\left\{-\frac{\partial}{\partial x_2}\left(-i\frac{\partial\psi_1}{\partial x_1}+\frac{\partial\psi_1}{\partial x_2}+i\frac{\partial\psi_2}{\partial x_3}\right)\right\}\\&+\frac{(\hbar c)^2}{2mc^2}\left\{-i\frac{\partial}{\partial x_3}\left(-i\frac{\partial\psi_2}{\partial x_1}-\frac{\partial\psi_2}{\partial x_2}-i\frac{\partial\psi_1}{\partial x_3}\right)\right\}\end{aligned}$$

$$\left(E - mc^2\right)\psi_1 = \frac{\hbar^2}{2m}\left\{-\frac{\partial^2\psi_1}{\partial x_1^2} - i\frac{\partial^2\psi_1}{\partial x_1\partial x_2} + \frac{\partial^2\psi_2}{\partial x_1\partial x_3}\right\}$$
$$+\frac{\hbar^2}{2m}\left\{i\frac{\partial^2\psi_1}{\partial x_1\partial x_2} - \frac{\partial^2\psi_1}{\partial x_2^2} - i\frac{\partial^2\psi_2}{\partial x_2\partial x_3}\right\}$$
$$+\frac{\hbar^2}{2m}\left\{-\frac{\partial^2\psi_2}{\partial x_1\partial x_3} - i\frac{\partial^2\psi_2}{\partial x_2\partial x_3} - \frac{\partial^2\psi_1}{\partial x_3^2}\right\}$$

$$\left(E - mc^2\right)\psi_1 = -\frac{\hbar^2}{2m}\nabla^2\psi_1$$

Likewise the second row becomes

$$\left(E - mc^2\right)\psi_2 = -\frac{\hbar^2}{2m}\nabla^2\psi_2$$

The third and fourth rows can be neglected in the non relativistic limit e.g. ψ_3 becomes

$$2mc^2\psi_3 = \hbar c\left\{\left(\ -i\tfrac{\partial\psi_2}{\partial x_1}\ \right) + \left(\ -\tfrac{\partial\psi_2}{\partial x_2}\ \right) + \left(\ -i\tfrac{\partial\psi_1}{\partial x_3}\ \right)\right\}$$

$$\psi_3 = \frac{\hbar}{2mc}\left\{\left(\ -i\tfrac{\partial\psi_2}{\partial x_1}\ \right) + \left(\ -\tfrac{\partial\psi_2}{\partial x_2}\ \right) + \left(\ -i\tfrac{\partial\psi_1}{\partial x_3}\ \right)\right\}$$

With $E' = \left(E - mc^2\right)$, in the non relativistic limit we have the equation for a free electron

$$E'\psi_1 = -\frac{\hbar^2}{2m}\nabla^2\psi_1$$

$$E'\psi_2 = -\frac{\hbar^2}{2m}\nabla^2\psi_2$$

and we have the Pauli spinor below corresponding to 2 identical Schrödinger equations, one for each spin component.

$$\phi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$$

- The Dirac spinor has four components.
- In the non-relativistic limit, the lower two are negligible to leading order.
- The upper two remain and form a two-component Pauli spinor.
- Those two components correspond to spin-up and spin-down.
- Therefore the non-relativistic limit naturally yields two independent Schrödinger equations, one for each spin state.

3 Spin

3.1 Magnetic Term (Static Field)

The classical section on Weber/Ampere is optional background. Its only purpose is to motivate why a term $e\dot{\mathbf{x}} \cdot \mathbf{A}$ must appear in the Lagrangian.

3.1.1 Lorentz Force

We normally see the force expressed in terms of the magnetic field $\vec{B}$

$$d\vec{F}^{A}_{V\text{ on }I_i d\vec{l}_i} = I_i d\vec{l}_i \times \left(\frac{\mu_0}{4\pi} \iiint_{V_j} \left(\vec{j}\left(\vec{r}_j, t\right) \times \frac{\vec{r_{ij}}}{r_{ij}^3} \right) \right) dV$$

$$d\vec{F}^{A}_{V\text{ on }I_i d\vec{l}_i} = I_i d\vec{l}_i \times \vec{B}\left(\vec{r}_i, t\right)$$

$$I_i d\vec{l}_i = dq_i \left(\vec{v}_{i+} - \vec{v}_{i-}\right) = dq_i \vec{v}_{i+,i-}$$

$$d\vec{F}^{A}_{V\text{ on }dq_i} = e\vec{v}_{i+,i-} \times \vec{B}\left(\vec{r}_i, t\right)$$

3.1.2 Ampere's Force Law

The magnetic force between current elements can also be derived directly from Ampere's force law which is given below in vector notation

$$d^2\vec{F}^{A}_{ji} = -\frac{\mu_0}{4\pi} I_i I_j \frac{\hat{r}_{ij}}{r_{ij}^2} \left[2\left(d\vec{l}_i \cdot d\vec{l}_j\right) - 3\left(\hat{r}_{ij} \cdot d\vec{l}_i\right)\left(\hat{r}_{ij} \cdot d\vec{l}_j\right) \right]$$

This leads to Assis **Eq 4.50**.

$$d\vec{F}_{C_j\text{on }I_i dl_i} = \frac{\mu_0}{4\pi} I_i I_j dl_i \oint_{C_j} \frac{1}{r_{ij}^3} \left[\hat{x}\left(x_j dz_j - z_j dx_j\right) + \hat{y}\left(y_j dz_j - z_j dy_j\right)\right]$$

There is no "magnetic field" as such in the above derivation.

The equivalence of this to the usual Lorentz form is shown below. Following Assis, I will have element $d\vec{l}_i$ at the origin so $\vec{r}_{ij} = -\hat{x}x_j - \hat{y}y_j - \hat{z}z_j$ and we have a minus sign here $-\frac{\mu_0}{4\pi}$

$$d\vec{B}\left(\vec{r}\right) = \frac{\mu_0}{4\pi} \left(I_j d\vec{l}_j \times \frac{\vec{r_{ij}}}{r_{ij}^3} \right) = -\frac{\mu_0}{4\pi} \frac{I_j}{r_{ij}^3} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ dx_j & dy_j & dz_j \\ x_j & y_j & z_j \end{vmatrix}$$

$$= -\frac{\mu_0}{4\pi} \left(\frac{I_j}{r_{ij}^3} \right) \left[\hat{x}\left(dy_j z_j - dz_j y_j\right) + \hat{y}\left(dz_j x_j - dx_j z_j\right) + \hat{z}\left(dx_j y_j - dy_j x_j\right)\right]$$

With $d\vec{l}_i$ aligned with the z axis we then have the same result as can be got from Ampere's force law.

$$I_i d\vec{l}_i \times \frac{\mu_0}{4\pi}\left(I_j d\vec{l}_j \times \frac{\vec{r_{ij}}}{r_{ij}^3}\right) = -\frac{\mu_0}{4\pi}\frac{I_i I_j}{r_{ij}^3}\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & dz_i \\ (dy_j z_j - dz_j y_j) & (dz_j x_j - dx_j z_j) & (dx_j y_j - dy_j x_j) \end{vmatrix}$$

$$= \frac{\mu_0}{4\pi} I_i I_j dl_i \frac{1}{r_{ij}^3}\left[\hat{x}\left(x_j dz_j - z_j dx_j\right) + \hat{y}\left(y_j dz_j - z_j dy_j\right)\right]$$

3.1.3 Weber's Force

Weber force between 2 charges is given by

$$d^2\vec{F}_{ji} = \frac{dq_i dq_j}{4\pi\epsilon_0}\frac{\hat{r}_{ij}}{r_{ij}^2}\left[1 + \frac{1}{c^2}\left(\vec{v}_{ij}\cdot\vec{v}_{ij} - \frac{3}{2}\left(\hat{r}_{ij}\cdot\vec{v}_{ij}\right)^2 + \vec{r}_{ij}\cdot\vec{a}_{ij}\right)\right]$$

The total force on current elements formed of charges dq_{i+}, dq_{i-} and dq_{j+}, dq_{j-} is then just the usual terms $[i+, j+], [i+, j-]\,[i-, j+]\,[i-, j-]$ i.e.

$$d^2\vec{F} = d^2\vec{F}_{i+,j+} + d^2\vec{F}_{i+,j-} + d^2\vec{F}_{i-,j+} + d^2\vec{F}_{i-,j-}$$

The derivation leads to the Assis **Eq 4.22**

$$d^2\vec{F}_{ji} = -\frac{dq_i dq_j}{4\pi\epsilon_0}\frac{\hat{r}_{ij}}{r_{ij}^2}\left[2\frac{(\vec{v}_{i+} - \vec{v}_{i-})\cdot(\vec{v}_{j+} - \vec{v}_{j-})}{c^2} - 3\frac{\left[\hat{r}_{ij}\cdot(\vec{v}_{i+} - \vec{v}_{i-})\right]\left[\hat{r}_{ij}\cdot(\vec{v}_{j+} - \vec{v}_{j-})\right]}{c^2}\right]$$

Note that although Weber's force law is based on relative motion between the charges, we see that *both* charges will need to be in motion if we are to have a force i.e. $(\vec{v}_{i+} - \vec{v}_{i-})$ and $(\vec{v}_{j+} - \vec{v}_{j-})$ need to be non zero.

3.1.4 Weber Potential Energy

Franz Neumann introduced the magnetic potential $\vec{A}(\vec{r_i})$ at circuit C_i due to circuit C_j

$$\vec{A}(\vec{r_i}) \equiv \frac{\mu_0}{4\pi}\oint_{C_j}\left(I_j\frac{d\vec{l}_j}{r_{ij}}\right) \tag{10}$$

The Weber potential energy of 2 charges i, j at a distance r_{ij} and a relative velocity v_{ij} is given below. (Weber's force law can be derived from this, see [Weber's Electrodynamics] and [Weber Study Notes])

$$d^2U = \frac{dq_i dq_j}{4\pi\epsilon_0}\frac{1}{r_{ij}}\left(1 - \frac{v_{ij}^2}{2c^2}\right)$$

$$d^2U = \frac{dq_{i+} dq_{j+}}{4\pi\epsilon_0 c^2}\frac{1}{r_{ij}}\left(\hat{r_{ij}}\cdot(\vec{v}_{i+} - \vec{v}_{i-})\right)\left(\hat{r_{ij}}\cdot(\vec{v}_{j+} - \vec{v}_{j-})\right)$$

Moving to current elements gives

$$U = \frac{\mu_0}{4\pi} I_i I_j \oint_{C_i} \oint_{C_j} \frac{\left(\hat{r_{ij}} \cdot \vec{dl_i}\right)\left(\hat{r_{ij}} \cdot \vec{dl_j}\right)}{r_{ij}}$$

$$U = \frac{\mu_0}{4\pi} I_i I_j \left[\oint_{C_i} \oint_{C_j} \left(\frac{\vec{dl_i} \cdot \vec{dl_j}}{r_{ij}}\right)\right]$$

$$U = I_i \oint_{C_i} \left[\frac{\mu_0}{4\pi} \oint_{C_j} I_j \left(\frac{\vec{dl_j}}{r_{ij}}\right)\right] \cdot \vec{dl_i}$$

$$U = I_i \oint_{C_i} \vec{A}\left(\vec{r_i}\right) \cdot \vec{dl_i}$$

Going from current elements back to charges

$$dU = dq_i \vec{v_i} \cdot \vec{A}\left(\vec{r_i}\right) = e\dot{\mathbf{x}} \cdot \mathbf{A}$$

3.1.5 Classical Lagrangian

For a free particle with no forces the Lagrangian is the kinetic energy $L = \frac{1}{2}m\dot{x}^2$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_i}\right) - \left(\frac{\partial L}{\partial x_i}\right) = 0$$

We need to modify the Lagrangian to give the magnetic force above. Let us try

$$L = \frac{1}{2}m\dot{x}^2 + e\dot{\mathbf{x}} \cdot \mathbf{A}$$

and look at the $e\dot{\mathbf{x}} \cdot \mathbf{A}$ term.

3.1.6 Classical Momentum Term

We get momentum by applying $\frac{\partial}{\partial \dot{x}_i}$ to the Lagrangian. i ranges from 1 to 3.

$e\dot{x}_i A_i$

$$\frac{\partial L}{\partial \dot{x}_i} = m\dot{x}_i + eA_i$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} = m\ddot{x} + e\frac{dA_i}{dt}$$

$\frac{\partial A_i}{\partial t} = 0$ i.e. A_i is only changing due to position for a static magnetic field.

$$e\frac{dA_i}{dt} = e\frac{\partial A_i}{\partial x_j}\frac{dx_j}{dt} = e\dot{x}_j\frac{\partial A_i}{\partial x_j}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} = m\ddot{x} + e\dot{x}_j\frac{\partial A_i}{\partial x_j}$$

3.1.7 Position Term

$e\dot{x}_j A_j$

$$\frac{\partial L}{\partial x_i} = e\frac{\partial\left(\dot{x}_j A_j\right)}{\partial x_i} = e\dot{x}_j\frac{\partial A_j}{\partial x_i}$$

3.1.8 Both Terms

From $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x_i}}\right) - \left(\frac{\partial L}{\partial x_i}\right) = 0$

$$m\ddot{x} + e\left(\dot{x}_j\frac{\partial A_i}{\partial x_j} - \dot{x}_j\frac{\partial A_j}{\partial x_i}\right) = 0$$

$$m\ddot{x} = e\left(\dot{x}_j\frac{\partial A_j}{\partial x_i} - \dot{x}_j\frac{\partial A_i}{\partial x_j}\right)$$

To see another way this can be represented, let us calculate $\mathbf{v}\times(\nabla\times\mathbf{A})$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$\hat{i}\left(\frac{\partial A_3}{\partial x_2} - \frac{\partial A_2}{\partial x_3}\right) - \hat{j}\left(\frac{\partial A_3}{\partial x_1} - \frac{\partial A_1}{\partial x_3}\right) + \hat{k}\left(\frac{\partial A_2}{\partial x_1} - \frac{\partial A_1}{\partial x_2}\right)$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \dot{x}_1 & \dot{x}_2 & \dot{x}_3 \\ \left(\frac{\partial A_3}{\partial x_2} - \frac{\partial A_2}{\partial x_3}\right) & -\left(\frac{\partial A_3}{\partial x_1} - \frac{\partial A_1}{\partial x_3}\right) & \left(\frac{\partial A_2}{\partial x_1} - \frac{\partial A_1}{\partial x_2}\right) \end{vmatrix}$$

The $\hat{i}$ component is

$$\dot{x}_2\left(\frac{\partial A_2}{\partial x_1} - \frac{\partial A_1}{\partial x_2}\right) + \dot{x}_3\left(\frac{\partial A_3}{\partial x_1} - \frac{\partial A_1}{\partial x_3}\right)$$

$$= -\dot{x}_2\frac{\partial A_1}{\partial x_2} - \dot{x}_3\frac{\partial A_1}{\partial x_3} + \dot{x}_2\frac{\partial A_2}{\partial x_1} + \dot{x}_3\frac{\partial A_3}{\partial x_1}$$

If we calculate the $\hat{i}$ component of $\left(\dot{x}_j\frac{\partial A_j}{\partial x_i} - \dot{x}_j\frac{\partial A_i}{\partial x_j}\right)$ we get the same.

$$\dot{x}_1\frac{\partial A_1}{\partial x_1} + \dot{x}_2\frac{\partial A_2}{\partial x_1} + \dot{x}_3\frac{\partial A_3}{\partial x_1} - \dot{x}_1\frac{\partial A_1}{\partial x_1} - \dot{x}_2\frac{\partial A_1}{\partial x_2} - \dot{x}_3\frac{\partial A_1}{\partial x_3}$$

$$= -\dot{x}_2\frac{\partial A_1}{\partial x_2} - \dot{x}_3\frac{\partial A_1}{\partial x_3} + +\dot{x}_2\frac{\partial A_2}{\partial x_1} + \dot{x}_3\frac{\partial A_3}{\partial x_1}$$

So $F = e\mathbf{v}\times(\nabla\times\mathbf{A})$ or

$$m\ddot{x} = e\mathbf{v}\times\mathbf{B}$$

3.2 Spin I

Originally with no magnetic field

$$\left[\gamma_\mu \frac{\partial}{\partial x_\mu} + M\right]\psi = 0$$

From the last section we added $e\dot{\mathbf{x}}\cdot\mathbf{A}$ to the Lagrangian to account for the force due to a static magnetic field i.e. $L = \frac{1}{2}m\dot{x}_k^2 + e\dot{x}_k A_k$. To get momentum from the Lagrangian we use $\frac{\partial}{\partial \dot{x}_k}$ so $p_k = m\dot{x}_k + eA_k$ and the quantum mechanical operator is now

$$\hat{p} = -i\hbar\frac{\partial}{\partial x_k} + eA_k = -i\hbar\left(\frac{\partial}{\partial x_k} - \frac{e}{i\hbar}A_k\right) = -i\hbar\left(\frac{\partial}{\partial x_k} + \frac{ie}{\hbar}A_k\right)$$

As in section 2.1 , replacing the classical energy and momentum in $E^2 = m^2c^4 + c^2p^2$is now given by

$$\left[-\hbar^2\frac{\partial^2}{\partial t^2} - m^2c^4 + c^2\hbar^2\left(\frac{\partial}{\partial x_k} + \frac{ie}{\hbar}A_k\right)^2\right]\psi = 0$$

and using $M = \frac{mc}{\hbar}$

$$\left[\left(\frac{\partial}{\partial x_k} + \frac{ie}{\hbar}A_k\right)^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - M^2\right]\psi = 0$$

which reduces to 6 if there is no magnetic field.

This becomes using the 4 vector notation - even though $A_0 = 0$ for a static magnetic field.

$$\left[\gamma_\mu\left(\frac{\partial}{\partial x_\mu} + \frac{ie}{\hbar}A_\mu\right) + M\right]\psi = 0 \tag{11}$$

3.3 Spin II

$$\left[\gamma_\lambda\left(\frac{\partial}{\partial x_\lambda} + \frac{ie}{\hbar}A_\lambda\right) - M\right]\left[\gamma_\mu\left(\frac{\partial}{\partial x_\mu} + \frac{ie}{\hbar}A_\mu\right) + M\right]\psi = 0$$

$$\left[\gamma_\lambda\gamma_\mu\left(\frac{\partial}{\partial x_\lambda} + \frac{ie}{\hbar}A_\lambda\right)\left(\frac{\partial}{\partial x_\mu} + \frac{ie}{\hbar}A_\mu\right) - M^2\right]\psi = 0$$

For $\lambda = \mu$

$$\left(\frac{\partial}{\partial x_\mu} + \frac{ie}{\hbar}A_\mu\right)^2 - M^2$$

For $\lambda \neq \mu$

$$\gamma_\lambda\gamma_\mu\left(\frac{\partial}{\partial x_\lambda} + \frac{ie}{\hbar}A_\lambda\right)\left(\frac{\partial}{\partial x_\mu} + \frac{ie}{\hbar}A_\mu\right)$$

3.4 Spin III

The key thing is that $[D_0, D_1] \neq 0$ because we now have $\left(\frac{\partial}{\partial x_\lambda} + \frac{ie}{\hbar}A_\lambda\right)$ instead of $\left(\frac{\partial}{\partial x_\lambda}\right)$

For $\lambda = 0$, $\mu = 1$ and $\lambda = 1$, $\mu = 0$
$\gamma_0\gamma_1 D_0 D_1 + \gamma_1\gamma_0 D_1 D_0 = \gamma_0\gamma_1 D_0 D_1 + (-\gamma_0\gamma_1) D_1 D_0 = \gamma_0\gamma_1(D_0 D_1 - D_1 D_0)$.

$$\gamma_0\gamma_1 D_0 D_1 = \gamma_0\gamma_1\left(\partial_0 + \frac{ie}{\hbar}A_0\right)\left(\partial_1 + \frac{ie}{\hbar}A_1\right)$$

$$\gamma_1\gamma_0 D_1 D_0 = \gamma_1\gamma_0\left(\partial_1 + \frac{ie}{\hbar}A_1\right)\left(\partial_0 + \frac{ie}{\hbar}A_0\right)$$

$$= \gamma_0\gamma_1\left[\partial_0\partial_1 + \left(\frac{ie}{\hbar}\right)\partial_0 A_1 + \frac{ie}{\hbar}A_0\partial_1 - \left(\frac{ie}{\hbar}\right)^2 A_0 A_1\right]$$
$$- \gamma_0\gamma_1\left[\partial_1\partial_0 + \left(\frac{ie}{\hbar}\right)\partial_1 A_0 + \frac{ie}{\hbar}A_1\partial_0 - \left(\frac{ie}{\hbar}\right)^2 A_1 A_0\right]$$

$$= \gamma_0\gamma_1\left[\frac{ie}{\hbar}\left(\partial_0 A_1 - \partial_1 A_0\right) + \frac{ie}{\hbar}\left(A_0\partial_1 - A_1\partial_0\right)\right]$$

Mandl does not mention the operator term so we are left with

$$= \gamma_0\gamma_1\left[\frac{ie}{\hbar}\left(\frac{\partial A_1}{\partial x_0} - \frac{\partial A_0}{\partial x_1}\right)\right]$$

For all $\lambda \neq \mu$

$$= \gamma_\lambda\gamma_\mu\frac{ie}{\hbar}\left(\nabla \times \mathbf{A}\right) = \frac{e}{\hbar}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B}$$

3.5 Spin IV

Putting together the terms where $\lambda \neq \mu$, $\frac{e}{\hbar}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B}$ and $\lambda = \mu$, $\left[\left(\frac{\partial}{\partial x_\mu} + \frac{ie}{\hbar}A_\mu\right)^2 - M^2\right]$.

$$\left[\left(\frac{\partial}{\partial x_\mu} + \frac{ie}{\hbar}A_\mu\right)^2 - M^2 + \frac{e}{\hbar}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B}\right]\psi = 0$$

Because of time independence of the magnetic field we have $A_0 = 0$ so $\left(\frac{\partial}{\partial x_0} + \frac{ie}{\hbar}A_0\right)^2 = \left(\frac{\partial}{\partial x_0}\right)^2$

$$\left(\frac{\partial}{\partial x_0} - M\right)\left(\frac{\partial}{\partial x_0} + M\right)\psi = \left[-\sum_k\left(\frac{\partial}{\partial x_k} + \frac{ie}{\hbar}A_k\right)^2 - \frac{e}{\hbar}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B}\right]\psi$$

$$\left(\frac{\partial}{\partial x_0} - M\right)\left(\frac{\partial}{\partial x_0} + M\right)\psi = [...]\,\psi$$

$$\left(\frac{i}{c}\frac{\partial}{\partial t} - \frac{mc}{\hbar}\right)\left(\frac{i}{c}\frac{\partial}{\partial t} + \frac{mc}{\hbar}\right)\psi = [...]\,\psi$$

$$\left(i\hbar\frac{\partial}{\partial t} - mc^2\right)\left(i\hbar\frac{\partial}{\partial t} + mc^2\right)\psi = \hbar^2 c^2\,[...]\,\psi$$

$i\hbar\frac{\partial\psi}{\partial t} = E\psi$. At non relativistic velocity $\left(i\hbar\frac{\partial}{\partial t}\right)\psi = \left(mc^2\right)\psi$. Note I am assuming the $+E$ solution of the Dirac equation here.

$$\left(i\hbar\frac{\partial}{\partial t} - mc^2\right)\left(2mc^2\right)\psi = \hbar^2 c^2\,[...]\,\psi$$

$$\left(i\hbar\frac{\partial}{\partial t} - mc^2\right)\psi = \frac{\hbar^2}{2m}\,[...]\,\psi$$

$$i\hbar\frac{\partial\psi}{\partial t} = \left[mc^2 + \frac{\hbar^2}{2m}\,[...]\right]\psi$$

$$i\hbar\frac{\partial\psi}{\partial t} = \left[mc^2 - \frac{\hbar^2}{2m}\sum_{k=1}^{3}\left(\frac{\partial}{\partial x_k} + \frac{ie}{\hbar}A_k\right)^2 - \frac{e\hbar}{2m}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B}\right]\psi \qquad (12)$$

We see the $\frac{e\hbar}{2m}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B}$ term which is the intrinsic *spin*-magnetic interaction.

3.6 Z Direction

Let us assume $\mathbf{A} = (0, Bx_1, 0)$ i.e. $A_2 = Bx_1$ so $\mathbf{A}$ is in the y direction and the magnitude of $\mathbf{A}$ increases as we move along the x direction.

From $\nabla \times \mathbf{A}$ this gives a magnetic field in the z direction.

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ 0 & Bx_1 & 0 \end{vmatrix} = \frac{\partial A_2}{\partial x_1}\hat{k} = \frac{\partial\left(Bx_1\right)}{\partial x_1}\hat{k} = B\hat{k}$$

i.e. $\mu = 2$ and $\lambda = 1$ when using $\frac{e\hbar}{2m}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B} = \frac{e\hbar}{2m}\left(i\gamma_\lambda\gamma_\mu\right)\left[\frac{\partial A_\mu}{\partial x_\lambda} - \frac{\partial A_\lambda}{\partial x_\mu}\right]$

$i\gamma_1\gamma_2$ gives the 1's across the diagonals corresponding to spin up and spin down.

$$i\gamma_1\gamma_2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} = i\begin{pmatrix} i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Eq 12 becomes

$$i\hbar\frac{\partial}{\partial t}\begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\\ \psi_4\end{pmatrix} = \left[mc^2 - \frac{\hbar^2}{2m}\sum_{k=1}^{3}\left(\frac{\partial}{\partial x_k}+\frac{ie}{\hbar}A_k\right)^2 - \frac{e\hbar}{2m}\begin{pmatrix}-1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & -1 & 0\\ 0 & 0 & 0 & 1\end{pmatrix}\mathbf{B}\right]\begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\\ \psi_4\end{pmatrix}$$

and

$$i\hbar\frac{\partial\psi_1}{\partial t} = \left[mc^2 - \frac{\hbar^2}{2m}\sum_{k=1}^{3}\left(\frac{\partial}{\partial x_k}+\frac{ie}{\hbar}A_k\right)^2 + \frac{e\hbar}{2m}\mathbf{B}\right]\psi_1$$

$$i\hbar\frac{\partial\psi_2}{\partial t} = \left[mc^2 - \frac{\hbar^2}{2m}\sum_{k=1}^{3}\left(\frac{\partial}{\partial x_k}+\frac{ie}{\hbar}A_k\right)^2 - \frac{e\hbar}{2m}\mathbf{B}\right]\psi_2$$

The same pair of equations holds for ψ_3 and ψ_4, but in the non-relativistic, positive-energy limit the lower components can be neglected. We therefore keep only ψ_1 and ψ_2 which form the Pauli spinor.

We have the terms

$$(\pm 1)\left(\frac{e}{m_e}\right)\left(\frac{\hbar}{2}\right)B$$

The result resembles the classical result of a spinning electron except for a factor 2 but we have arrived at this result without reference to a spinning electron.

3.7 Quantum Spin

$$\mu = \left(\frac{e}{m_e}\right)\left(\frac{\hbar}{2}\right) = \left(\frac{e}{m_e}\right)S$$

3.8 Classical Spin

Classically the magnetic moment of a spinning charged shell is half of the above

$$\mu = \frac{1}{2}\left(\frac{e}{m_e}\right)S$$

This is derived as below

Assume a shell of radius a which charge e uniformly distributed over the surface. The surface charge density is

$$\sigma = \frac{e}{4\pi a^2}$$

The moment of inertia of a shell is

$$I = \frac{2}{3}m_e a^2$$

The angular momentum is

$$S = I\omega = \frac{2}{3} m_e a^2 \omega$$

Using co-latitude the charge on a ring between θ and $\theta d\theta$ is

$$dq = \sigma 2\pi \left(a \sin\theta\right)\left(a d\theta\right) = \left(\frac{e}{4\pi a^2}\right) 2\pi \left(a \sin\theta\right)\left(a d\theta\right) = \frac{e}{2} \sin\theta d\theta$$

The ring rotates with ω i.e. with frequency $f = \frac{\omega}{2\pi}$ so effective current

$$dI = f dq = \frac{\omega}{2\pi}\left(\frac{e}{2}\sin\theta d\theta\right) = \frac{e\omega}{4\pi}\sin\theta d\theta$$

The magnetic moment is $\mu = IA$ so

$$d\mu = \left(\frac{e\omega}{4\pi}\sin\theta d\theta\right)\pi a^2 \sin^2\theta = \frac{e\omega a^2}{4}\sin^3\theta d\theta$$

$\int_0^\pi \sin^3\theta d\theta = \frac{4}{3}$

$$\mu = \frac{e\omega a^2}{4}\frac{4}{3} = \frac{e\omega a^2}{3}$$

From the angular momentum $S = \frac{2}{3} m_e a^2 \omega$

$$\left(\frac{e}{2m_e}\right) S = \left(\frac{e}{2m_e}\right)\left(\frac{2}{3} m_e a^2 \omega\right) = \frac{e\omega a^2}{3} = \mu$$

so

$$\mu = \frac{1}{2}\left(\frac{e}{m_e}\right) S$$

3.9 Summary

The start of this section is the formula for a moving charge in a static magnetic field. This force can be got from Weber's force law between 2 charges.

This leads to an extra term in the Lagrangian ($e\dot{\mathbf{x}} \cdot \mathbf{A}$) which is needed to account for this force.

But that extra term implies a "momentum" if we apply the derivative $\frac{\partial}{\partial \dot{x}}$ to the Lagrangian.

We then substitute $-i\hbar\frac{\partial}{\partial x}$ for this momentum in the Dirac equation.

This then leads to $\frac{e\hbar}{2m}\left(i\gamma_\lambda\gamma_\mu\right)\mathbf{B}$

A Classical Potential Energy of Magnetic Moment

A magnetic dipole moment μ (e.g., from a current loop, a bar magnet, or a spinning electron) experiences a torque when placed in an external magnetic field B. The torque τ on the dipole is given by

$$\tau = \mu \times B$$

Work done is $W = \mathbf{F} \cdot \mathbf{d}$. The negative sign is because as we increase θ the torque is opposing that increase. Increasing θ is increasing the potential energy.

$$dW_{\text{field}} = -\tau d\theta = -\mu B \sin\theta d\theta$$

$$U = -\left[\int (-\mu B \sin\theta)\, d\theta\right] = -\mu B \cos\theta + C$$

Define $U = 0$ when the dipole is perpendicular i.e. $\theta = \frac{\pi}{2}$ so $C = 0$

$$U = -\mu B \cos\theta = -\mu \cdot \mathbf{B}$$

$$U_0 = -\mu B$$

$$U_\pi = \mu B$$

B Stern-Gerlach Experiment 1922

Grok explains this well. The experiment is based on Silver atoms $\left({}^{47}\text{Ag}\right)$ that have a single unpaired electron in the outer shell. Also because the orbital angular momentum quantum number for this state is $l = 0$ there is no orbital angular momentum and no orbital magnetic moment μ_L.

A beam of silver atoms is passed through a narrow gap between the poles of an *inhomogeneous* magnetic field.

By convention magnetic fields point from the North pole to the South pole. Assume the magnetic field is in the $+\hat{z}$ direction. A beam of silver atoms passes though the magnetic field but travel in a perpendicular direction (lets assume the $\hat{x}$ direction). The amount of time an atom travelling in the $\hat{x}$ direction at a typically velocity of 500m/s spends in the magnetic field of length $L = 0.01$m is

$$t = \frac{L}{v_x} = \frac{0.01\text{m}}{500\text{m/s}} = 2 \times 10^{-5}\text{s}$$

Classically a magnetic moment would align itself along the $+\hat{z}$ direction as this is the minimum potential energy.

From $F = -\frac{dU}{dz}$ and the above section A

$$F_z = -\frac{dU}{dz} = \frac{d}{dz}\left(\mu \cdot \mathbf{B}\right)$$

Quantum mechanically, spin and therefore μ_z i s already aligned with the z axis, and therefore magnetic field, via $S = \pm\frac{\hbar}{2}$ so there is no torque. Instead from the inhomogeneous magnetic field we have a translational force

$$F_z = \mu_z \frac{dB}{dz}$$

The magnet in the Stern-Gerlach experiment was designed to have a strong gradient

$$\frac{dB}{dz} = 10^3 \mathrm{T/m}$$

From the Dirac equation

$$\mu_z = \frac{e\hbar}{2m_e} \approx 9.274 \times 10^{-24} \mathrm{J/T}$$

$$F_z = \left(9.274 \times 10^{-24}\mathrm{J/T}\right)\left(10^3\mathrm{T/m}\right) \approx 9.274 \times 10^{-21}\mathrm{N}$$

This force is small, but it acts on a silver atom, which has a mass of approximately 1.8×10^{-25}kg

$$a_z = \frac{F_z}{m} = \frac{9.274 \times 10^{-21}\mathrm{N}}{1.8 \times 10^{-25}\mathrm{kg}} \approx 5.15 \times 10^4 \mathrm{m/s}$$

From $s = \frac{1}{2}a_z t^2$ the deflection of the silver atom while in the magnetic field is

$$s_z = \frac{1}{2}\left(5.15 \times 10^4\mathrm{m/s}\right)\left(2 \times 10^{-5}\mathrm{s}\right)^2 \approx 0.01\mathrm{mm}$$

Classically, the magnetic moment due to spin could be orientated in any direction so we'd expect the beam to be spread out across the detecting screen. However only 2 distinct spots were observed.

C Pauli Matrices

The Pauli matrices are listed again for reference

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These 3 matrices plus the identity matrix $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ form a complete orthonormal basis for 2x2 Hermitian vectors i.e. Every Hermitian 2×2 matrix H can be uniquely written as

$$H = a_0 I + a_x\sigma_x + a_y\sigma_y + a_z\sigma_z$$

with real coefficients a_μ

e.g. σ_x and σ_y are orthogonal because their inner product (*the trace*) is zero, and this orthogonality guarantees that you cannot express one as a linear combination of the others.

C.1 Hermitian

Pauli matrices are Hermitian.

A matrix is Hermitian if it is equal to its conjugate transpose $A = A^{\dagger}$.

The conjugate transpose of a matrix is obtained by taking the transpose (swap rows and columns) and then taking the complex conjugate of each element.

e.g. $\sigma_y = \sigma_y^{\dagger} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

In quantum mechanics, Hermitian operators represent observables (quantities you can measure).

C.2 Inverse

Pauli matrices are also their own inverse.

(Note: A matrix generally can be Hermitian but not its own inverse - and vice versa. Also both and neither.)

If a matrix is diagonalizable and $\mathrm{A}^{\mathbf{2}} = \mathrm{I}$, then the eigenvalues must be ± 1.

C.2.1 Eigenvalues

$$\det\left(\sigma_y - \lambda I\right) = \det\begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} = \lambda^2 - 1 = 0$$

$$\lambda = \pm 1$$

C.2.2 Eigenvectors

For eigenvalue $\lambda = 1$. From

$$\left(\sigma_y - \lambda I\right) v = 0$$

$$\begin{pmatrix} -1 & (-i) \\ i & -1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$-a - ib = 0 \implies a = -ib$$

Choosing $b = 1$

$$v_{+1} = \begin{pmatrix} -i \\ 1 \end{pmatrix}$$

$\|v\| = v^{*}v = i\,(-i) + 1 = 2$ so the normalised vector is

$$v_{+1} = \frac{1}{\sqrt{2}}\begin{pmatrix} -i \\ 1 \end{pmatrix}$$

Similarly

$$v_{-1} = \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1 \end{pmatrix}$$

C.2.3 Diagonalization

If we have a matrix A that is diagonalizable then

$$A = PDP^{-1}$$

where D is a diagonal matrix containing the eigenvalues of A.
P has the eigenvectors of A as its columns.
This comes from $AP = PD$.
It makes solving matrix equations simpler. $Ax = B$ becomes $\left(PDP^{-1}\right) x = B$.
We can then make a change of variable $X = P^{-1}x$ to give $PDX = B$.
Multiplying both sides by P^{-1} gives $DX = P^{-1}B$ which is simpler to solve.

C.2.4 Diagonalization - Pauli Matrix

All Hermitian matrices are guaranteed to be diagonalizable with real eigenvalues.

$$\sigma_y = PDP^{-1}$$

where D are the eigenvalues

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and P are the eigenvectors as columns

$$P = \frac{1}{\sqrt{2}} \begin{pmatrix} -i & i \\ 1 & 1 \end{pmatrix}$$

C.3 Product

$$\sigma_y \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{1}$$

$$\sigma_x \sigma_y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = i\sigma_z$$

C.4 Commutation

The Pauli matrices correspond to angular momentum around the x , y and z axis respectively
$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
These satisfy

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$$

Orbital angular momentum was known to have the following commutation property prior to Pauli's matrices

$$[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$$

from e.g. $L_x = yp_z - zp_y$ and $[x_i, p_j] = i\hbar\delta_{ij}$

Pauli introduced the matrices σ_i and spin operator $S_i = \frac{\hbar}{2}\sigma_i$ such that

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k$$

Let us try $S_i = a\sigma_i$

$$[a\sigma_x, a\sigma_y] = a^2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - a^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = a^2 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - a^2 \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

$$= ia^2 2 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = i2a\,(a\sigma_z) \implies a = \frac{\hbar}{2}$$

$$\left[\frac{\hbar}{2}\sigma_x, \frac{\hbar}{2}\sigma_y\right] = i\hbar\left(\frac{\hbar}{2}\sigma_z\right)$$

C.5 Eigenvectors

All 3 Pauli matrices are Hermitian and have real eigenvalues ± 1 corresponding to spin up and down.

They also satisfy $\sigma_i^2 = I$

For σ_z the eigenvectors are

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

From $S_i = \frac{\hbar}{2}\sigma_i$

$$S_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Note these eigenvectors/operators are unlike the usual spatial wavefunctions we are familiar with from solving Schrödinger's equation. They are spinors that the spin operator/Pauli matrices operates on. Unlike orbital angular momentum which arises from $r \times p$, spin is an intrinsic property.

D Quaternions

Quaternions were discovered by William Hamilton in 1843.

D.1 Definition

Think of quaternions as an extension of 2-D complex numbers $a + \mathbf{i}b$. They are represented as

$$Q = W + X\mathbf{i} + Y\mathbf{j} + Z\mathbf{k}$$

where W, X, Y, Z are real numbers and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are imaginary units satisfying

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$$

and

$$\mathbf{ij} = \mathbf{k}$$

$$\mathbf{jk} = \mathbf{i}$$

$$\mathbf{ki} = \mathbf{j}$$

and anti-commutativity (e.g., $\mathbf{j}\,\mathbf{i} = -\mathbf{k}$).

D.2 3-D Rotation

Although Hamilton didn't create quaternions with 3-D rotations in mind, quaternions turn out to be a very convenient way of representing rotations in 3-D space. The unit quaternion can be written as below and represents a rotation by angle θ around axis $\hat{n}$.

$$q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(\hat{n}_x i + \hat{n}_y j + \hat{n}_z k)$$

and a vector v is rotated via

$$v^{'} = qvq^{-1}$$

where v is treated as a pure quaternion

$$v = 0 + v_x i + v_y j + v_z k$$

$v^{'}$ is also a pure quaternion.

$v^{'} = qvq^{-1}$ is simply the classical Rodrigues' rotation formula below written in quaternion form.

$$v' = v\cos\theta + (\hat{n} \times v)\sin\theta + \hat{n}(\hat{n} \cdot v)(1 - \cos\theta)$$

D.2.1 Intuition

A complex number $v = x + iy$ represents a 2-D vector. Multiplying by $e^{i\theta}$ rotates it in the plane i.e. $v' = e^{i\theta}v$, and multiplying by $e^{-i\theta}$ rotates it in the opposite direction. So complex multiplication is a 2-D rotation. Because complex numbers commute the sandwich does nothing $v' = e^{i\theta}ve^{-i\theta} = v$.

For quaternions, qv is a rotation in 4-D space. However qvq^{-1} is a simple 4-D rotation that fixes the 2-plane spanned by the scalar axis and the rotation axis of q. (Important qv and vq^{-1} alone do not do this. It's the sandwich qvq^{-1} that does this). Its orthogonal complement is another 2-plane, which corresponds to the directions perpendicular to the rotation axis. The pure vector subspace decomposes into the fixed axis direction and this perpendicular 2-plane, so the sandwich preserves all pure quaternions and acts on them as a 3-D rotation.

It maps pure quaternions to pure quaternions.

D.2.2 Example

As a simple example, let use rotate a vector $(1, 0, 0)$ i.e. in the x direction, around the y axis i.e. $\hat{n} = (0, 1, 0)$ by $\frac{\pi}{2}$. The quaternion representing this rotation is

$$q = \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)(0i + 1j + 0k) = \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)j$$

$$v = i$$

$$qv = i\cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)ji = i\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)k$$

$$qv = i\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)k$$

$$q^{-1} = \cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)j$$

$$qvq^{-1} = \left[i\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)k\right]\left[\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)j\right]$$

$$= i\cos^2\left(\frac{\pi}{4}\right) - \cos\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{4}\right)ij - \sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{4}\right)k + \sin^2\left(\frac{\pi}{4}\right)kj$$

$$= i\left[\cos^2\left(\frac{\pi}{4}\right) - \sin^2\left(\frac{\pi}{4}\right)\right] - \cos\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{4}\right)k - \sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{4}\right)k$$

$\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$$qvq^{-1} = 0 - 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)k$$

$$qvq^{-1} = -k$$

D.3 Matrix Representation

If we use the following, then we find that $i\sigma_x, i\sigma_y, i\sigma_z$ satisfy the same multiplication rules as the quaternion units $\mathbf{i}, \mathbf{j}, \mathbf{k}$ above

$i\sigma_x = s_x = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

$i\sigma_y = s_y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$i\sigma_z = s_z = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$

Using this, $Q = W + X\mathbf{i} + Y\mathbf{j} + Z\mathbf{k}$ becomes

$$Q = W\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + X\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} + Y\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + Z\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

$$= \begin{pmatrix} W & 0 \\ 0 & W \end{pmatrix} + \begin{pmatrix} 0 & Xi \\ Xi & 0 \end{pmatrix} + \begin{pmatrix} 0 & Y \\ -Y & 0 \end{pmatrix} + \begin{pmatrix} Zi & 0 \\ 0 & -Zi \end{pmatrix}$$

$$Q = \begin{pmatrix} (W + iZ) & (Y + iX) \\ (iX - Y) & (W - iZ) \end{pmatrix}$$

The vector quaternion becomes a 2x2 matrix

$$v = (0, v_x, v_y, v_z)$$

$$V = v_x\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} + v_y\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + v_z\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

$$V = \begin{pmatrix} (iv_z) & (v_y + v_x i) \\ (v_x i - v_y) & (-iv_z) \end{pmatrix}$$

Note V has no real diagonal components as expected. It only has imaginary diagonal components. The off diagonal components are both real and imaginary.

D.3.1 Matrix Rotation

The quaternion rotation matrix is now

$$V_{\text{rot}} = QVQ^{-1}$$

with

$$Q = \begin{pmatrix} (W + iZ) & (Y + iX) \\ (iX - Y) & (W - iZ) \end{pmatrix}$$

$$Q^{-1} = \begin{pmatrix} (W - iZ) & (-iX - Y) \\ (Y - iX) & (W + iZ) \end{pmatrix}$$

The following is a brute force calculation of QV

$$QV = \begin{pmatrix} (W+iZ) & (Y+iX) \\ (iX-Y) & (W-iZ) \end{pmatrix} \begin{pmatrix} (iv_z) & (v_y+v_x i) \\ (v_x i - v_y) & (-iv_z) \end{pmatrix}$$

$$= \begin{pmatrix} [(W+iZ)(iv_z)+(Y+iX)(v_x i - v_y)] & [(W+iZ)(v_y+v_x i)+(Y+iX)(-iv_z)] \\ [(iX-Y)(iv_z)+(W-iZ)(v_x i - v_y)] & [(iX-Y)(v_y+v_x i)+(W-iZ)(-iv_z)] \end{pmatrix}$$

$$= \begin{pmatrix} [iWv_z - Zv_z + iYv_x - Yv_y - Xv_x - iXv_y] & [Wv_y + iWv_x + iZv_y - Zv_x - iYv_z + Xv_z] \\ [-Xv_z - iYv_z + iWv_x - Wv_y + Zv_x + iZv_y] & [iXv_y - Xv_x - Yv_y - iYv_x - iWv_z - Zv_z] \end{pmatrix}$$

$$= \begin{pmatrix} [i(Wv_z + Yv_x - Xv_y) - (Xv_x + Yv_y + Zv_z)] & [i(Wv_x + Zv_y - Yv_z) + (Wv_y - Zv_x + Xv_z)] \\ [i(Wv_x - Yv_z + Zv_y) - (-Xv_z - Wv_y + Zv_x)] & [i(Xv_y - Yv_x - Wv_z) - (Xv_x + Yv_y + Zv_z)] \end{pmatrix}$$

Note if the axis of rotation is y then $X = Z = 0$. If the vector is in the x direction then $v_y = v_z = 0$ so from $(Xv_x + Yv_y + Zv_z)$ there will be no real diagonal elements.

To get the rotated vector from QVQ^{-1} we now need to multiply this by Q^{-1}

$$Q^{-1} = \begin{pmatrix} (W-iZ) & (-iX-Y) \\ (Y-iX) & (W+iZ) \end{pmatrix}$$

and V_{rot} will be of the same form which makes it easy to read off the rotated vector $(v_x\prime, v_y\prime, v_z\prime)$ i.e.

$$V = \begin{pmatrix} \left(iv_z^{'}\right) & \left(v_y^{'} + v_x^{'} i\right) \\ \left(v_x^{'} i - v_y^{'}\right) & \left(-iv_z^{'}\right) \end{pmatrix}$$

The diagonal elements will be purely imaginary e.g.

The top left element is

$$\begin{aligned} &(i(Wv_z + Yv_x - Xv_y) - (Xv_x + Yv_y + Zv_z))(W - iZ) \\ &+ (i(Wv_x + Zv_y - Yv_z) + (Wv_y - Zv_x + Xv_z))(Y - iX) \end{aligned}$$

Looking at the real terms

$$\begin{aligned} &+ Z(Wv_z + Yv_x - Xv_y) \\ &- W(Xv_x + Yv_y + Zv_z) \\ &+ X(Wv_x + Zv_y - Yv_z) \\ &+ Y(Wv_y - Zv_x + Xv_z) \end{aligned}$$

$$\begin{aligned} v_x&(YZ - XW + XW - YZ) \\ +v_y&(-XZ - WY + XZ + WY) \\ +v_z&(WZ - WZ - XY + XY) \\ &= 0 \end{aligned}$$

D.4 Rotations Using Pauli Matrices

A rotation by angle θ about the unit axis $\hat{n}$ is represented by the quaternion

$$q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(\hat{n}_x i + \hat{n}_y j + \hat{n}_z k)$$

using the shorthand $\vec{i} = (i, j, k)$,we can write

$$q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)\left(\hat{n}\cdot\vec{i}\right)$$

Using Pauli matrices (and keeping the dot-product notation $\hat{n}\cdot\sigma$ to save us writing $n_x\sigma_x + n_y\sigma_y + n_z\sigma_z$)

$$Q = \cos\left(\frac{\theta}{2}\right) I + i\sin\left(\frac{\theta}{2}\right)(\hat{n}\cdot\sigma)$$

Now define the Rotation Operator $U(\theta, \hat{n})$ as

$$U(\theta, \hat{n}) = \exp\left(\frac{i}{2}\theta\hat{n}\cdot\sigma\right) = \cos\left(\frac{\theta}{2}\hat{n}\cdot\sigma\right) + i\sin\left(\frac{\theta}{2}\hat{n}\cdot\sigma\right)$$

From

$$\cos\left(\frac{\theta}{2}\hat{n}\cdot\sigma\right) = I - \frac{1}{2!}\left(\frac{\theta}{2}\right)^2(\hat{n}\cdot\sigma)^2 + \frac{1}{4!}\left(\frac{\theta}{2}\right)^4(\hat{n}\cdot\sigma)^4 ... = I - \frac{1}{2!}\left(\frac{\theta}{2}\right)^2 I + \frac{1}{4!}\left(\frac{\theta}{2}\right)^4 I...$$

$$= I\left[1 - \frac{1}{2!}\left(\frac{\theta}{2}\right)^2 + \frac{1}{4!}\left(\frac{\theta}{2}\right)^4 ...\right] = I\cos\left(\frac{\theta}{2}\right)$$

Similarly because all odd powers of $(\hat{n}\cdot\sigma)^n$ simplify to $\hat{n}\cdot\sigma$

$$\sin\left(\frac{\theta}{2}\hat{n}\cdot\sigma\right) = (\hat{n}\cdot\sigma)\left[\left(\frac{\theta}{2}\right) - \frac{1}{3!}\left(\frac{\theta}{2}\right)^3 ..\right] = (\hat{n}\cdot\sigma)\sin\left(\frac{\theta}{2}\right)$$

$$U(\theta, \hat{n}) = \exp\left(\frac{i}{2}\theta\hat{n}\cdot\sigma\right) = \cos\left(\frac{\theta}{2}\right) I + i\sin\left(\frac{\theta}{2}\right)\hat{n}\cdot\sigma$$

E Gamma Matrices

In Weyl (chiral) representation

$$\gamma_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\gamma_1 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma_2 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma_3 = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix}$$

E.1 Anti Commutation

The 4 matrices $\gamma_0, \gamma_1, \gamma_2, \gamma_3$, all satisfy the commutation relation $\gamma_\lambda\gamma_\mu + \gamma_\mu\gamma_\lambda = 2\delta_{\lambda\gamma}$

For the γ matrices formed from the Pauli matrices

$$\gamma_j\gamma_k = \begin{pmatrix} 0 & -i\sigma_j \\ i\sigma_j & 0 \end{pmatrix} \begin{pmatrix} 0 & -i\sigma_k \\ i\sigma_k & 0 \end{pmatrix} = \begin{pmatrix} (\sigma_j\sigma_k) & 0 \\ 0 & (\sigma_j\sigma_k) \end{pmatrix}$$

$$\gamma_k\gamma_j = \begin{pmatrix} 0 & -i\sigma_k \\ i\sigma_k & 0 \end{pmatrix} \begin{pmatrix} 0 & -i\sigma_j \\ i\sigma_j & 0 \end{pmatrix} = \begin{pmatrix} (\sigma_k\sigma_j) & 0 \\ 0 & (\sigma_k\sigma_j) \end{pmatrix}$$

It follows from the commutative properties of the Pauli matrices

$$\gamma_j\gamma_k + \gamma_k\gamma_j = 0$$

$$\gamma_k\gamma_k + \gamma_k\gamma_k = 2$$

$$\gamma_j\gamma_k + \gamma_k\gamma_j = 2\delta_{jk}$$

γ_0 also anti commutes with $\gamma_1, \gamma_2, \gamma_3$

F γ_5 matrix

$\gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_0$

$$\gamma_1\gamma_2 = \begin{pmatrix} i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}$$

$$\gamma_1\gamma_2\gamma_3 = \begin{pmatrix} i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix} \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

$$\gamma_1\gamma_2\gamma_3\gamma_0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_0 = \begin{pmatrix} 0 & -I_2 \\ -I_2 & 0 \end{pmatrix}$$

G Gamma Matrix Cyclical Relationships

G.1 $\gamma_1\gamma_2$

$$\gamma_1\gamma_2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}$$

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$I_2 \otimes \sigma_3 = \begin{pmatrix} 1\cdot\sigma_3 & 0\cdot\sigma_3 \\ 0\cdot\sigma_3 & 1\cdot\sigma_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$i\,(I_2 \otimes \sigma_3) = \begin{pmatrix} i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}$$

or in terms of γ_5

$$\gamma_0\gamma_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\gamma_0\gamma_5\gamma_3 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix} = \begin{pmatrix} -i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \end{pmatrix}$$

$$\hat{\sigma_3} = i\gamma_0\gamma_5\gamma_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$-i\gamma_1\gamma_2 = -i \begin{pmatrix} i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\hat{\sigma_3} = -i\gamma_1\gamma_2$$

H 4-Vectors

In special relativity we express physical quantities as 4 vectors which have 3 spatial and 1 time component.

So for velocity we have

$$U^\mu = (c, \vec{v})$$

For momentum

$$P^\mu = mU^\mu = (mc, m\vec{v})$$

From $E = mc^2$, $\frac{E}{c} = mc$

$$P^\mu = \left(\frac{E}{c}, p_x, p_y, p_z\right)$$

The norm of this is invariant which leads to

$$E^2 = m^2c^4 + c^2p^2$$

References

[Quantum Mechanics] Mandl

[Weber's Electrodynamics] Andre Koch Torres Assis

[Weber Study Notes] `https://xetle.com/pdf/Weber.pdf`